

Name:	<u>ANSWERS</u>	Date: 21/5/20
	Year 9 Mathematics 2020 Test: PYTHAGORAS Time allowed: 50 minutes Calculators allowed Resources Allowed: • 1 piece of A4 paper notes, back and front, typed or handwritten • Pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters • Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. • Use blue or black pen only please. Use pencil only for diagrams.	Total Marks: /50 %

Multiple Choice: Circle the correct answer(s)

(15 marks)

- 1 A triangle is said to satisfy the rule $c^2 = a^2 + b^2$ for which special triangle?

(A) acute angled
(B) right angled
(C) obtuse angled
(D) none of these
- 2 The longest side of a right angled triangle is called the

(A) shortest side
(B) middle side
(C) hypotenuse
(D) none of these
- 3 Given that $c^2 = a^2 + b^2$ and $a = 3$, $b = 4$ what is the value of c ?

a) 25
b) 9
c) 5
d) 16
- 4 Pythagoras' theorem can be applied to

(A) acute angled triangles
(C) right angled triangles

(B) obtuse angled triangles
(D) any triangle
- 5 The hypotenuse of a right angled triangle is 10cm. If one side is 6cm, the third side is

a) 16cm
b) 8cm
c) 5cm
d) 100cm
- 6 If two sides of a right angled triangle are 9 cm and 12cm then the hypotenuse is

a) 30cm
b) 15cm
c) 81cm
d) 144cm

6

- 7 The Pythagorean result for a triangle ABC right angled at C is
 (A) $a^2 = b^2 + c^2$ (B) $b^2 = a^2 + c^2$ (C) $c^2 = a^2 + b^2$ (D) none of these
- 8 The hypotenuse of a right angled triangle is opposite to the
 (A) acute angle (B) right angle (C) obtuse angle (D) none of these
- 9 If two sides of a right angled triangle are 16 cm and 12cm then the hypotenuse is
 a) 18cm b) 144cm c) 256cm d) 20cm
- 10 In a triangle ABC right angled at C, the hypotenuse is named as
 (A) a (B) b (C) c (D) none of these
- 11 If two sides of a right angled triangle are 15 cm and 20cm then the hypotenuse is
 a) 25cm b) 40cm c) 35cm d) 26cm
- 12 If $n^2 = 2304$ then n equals
 (A) 38 (B) 42 (C) 48 (D) 52
- 13 11^2 equals
 a) 112 b) 132 c) 121 d) 144
- 14 2.7^2 equals
 a) 2.7×2.7 b) $\frac{27^2}{10^2}$ c) 7.29 d) All of these
- 15 $\sqrt{81}$ equals
 a) 9.0 b) 9^2 c) 9 d) $\frac{27}{3}$

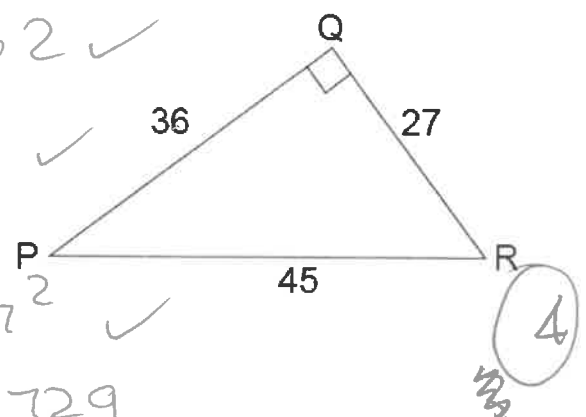
SHORT ANSWER

(35 marks)

1 If $n^2 = 3844$ then find the value of n 62 ✓

2 Is $\{6, 8, 10\}$ a Pythagorean triad? Yes ✓

3 Prove that $\triangle PQR$ is a right angled triangle.



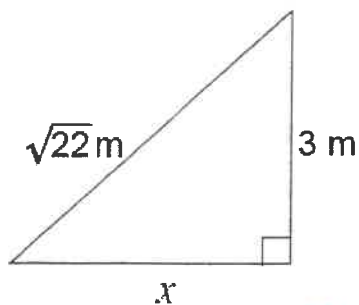
$$45^2 = 36^2 + 27^2 \quad \checkmark$$

$$2025 = 1296 + 729$$

$$2025 = 2025 \quad \checkmark$$

Find the length of the unknown side in the following triangles correct to two decimal places.

4



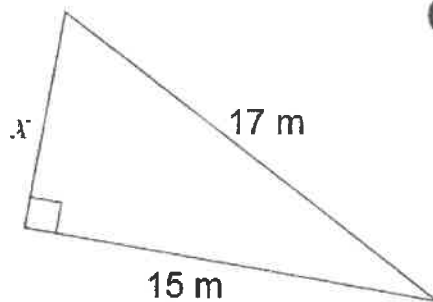
$$x^2 = (\sqrt{22})^2 - 3^2 \quad \checkmark$$

$$x^2 = 22 - 9$$

$$x = \sqrt{13}$$

$$x = 3.61 \text{ m} \quad \checkmark$$

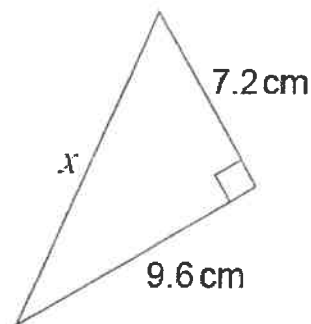
5



$$x^2 = 17^2 - 15^2 \quad \checkmark$$

$$x = 8.00 \text{ m} \quad \checkmark$$

6



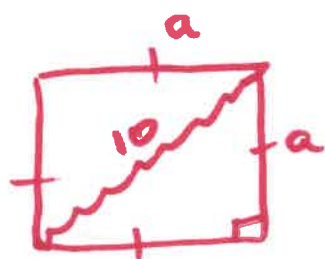
$$x^2 = 7.2^2 + 9.6^2 \quad \checkmark$$

$$x = 12.00 \text{ cm} \quad \checkmark$$

(6)

7

A square sheet of corrugated cardboard has a perforated diagonal measuring 10 cm in length. Find the length of the square (to 2 decimal places). First, draw and label a diagram.



$$a^2 + a^2 = 10^2 \quad \checkmark$$

$$2a^2 = 100 \quad \checkmark$$

$$a^2 = 50 \quad \checkmark$$

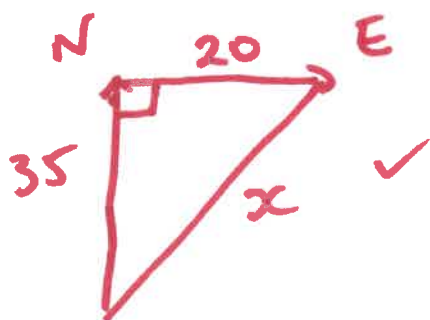
$$a = \sqrt{50}$$

$$a = 7.07 \text{ cm} \quad \checkmark$$

(5)

8

A ship leaves port and sails 35 km due North and then 20 km due East. If the ship returns to port by travelling along the diagonal route, how many km (to the nearest km) would the ship have sailed altogether? First, draw and label a diagram.



$$x^2 = 35^2 + 20^2 \quad \checkmark$$

$$x^2 = 1225 + 400$$

$$x^2 = 1625$$

$$x = 40.31 \quad \checkmark$$

(5)

$$\text{Total } 35 + 20 + 40 = 95 \text{ km} \quad \checkmark$$

7

9

If the area of square A is 144cm^2 and the area of square B is 256cm^2 . What is the length of side c ?

$$a = \sqrt{144} = 12 \checkmark$$

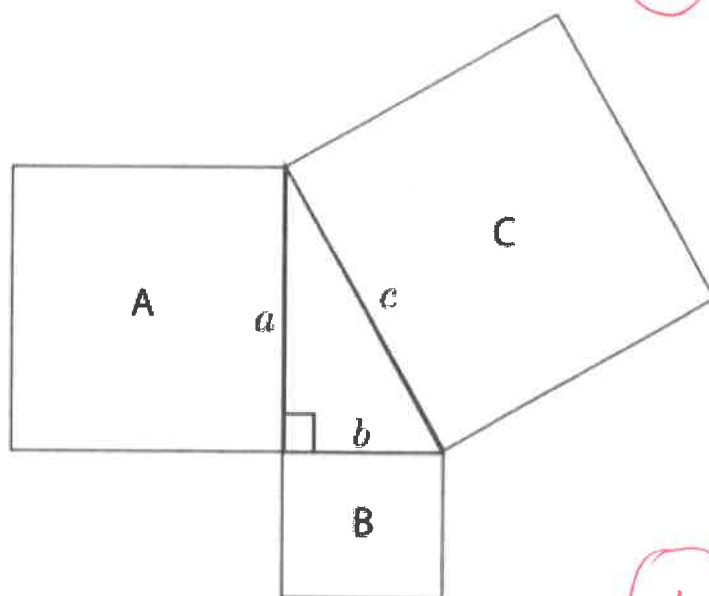
$$b = \sqrt{256} = 16 \checkmark$$

$$c^2 = 12^2 + 16^2$$

$$c = \sqrt{400}$$

$$c = 20 \text{ cm} \checkmark$$

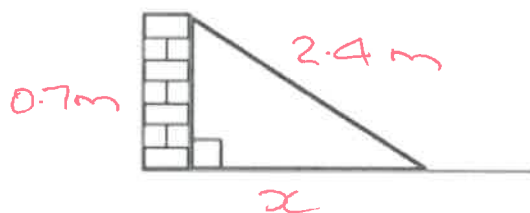
✓ units for whole test.



4

10

Kevin is building a ramp from the ground to the top of a brick veranda 0.7 m off the ground.



✓

1

a) Label the diagram with all known measurements.

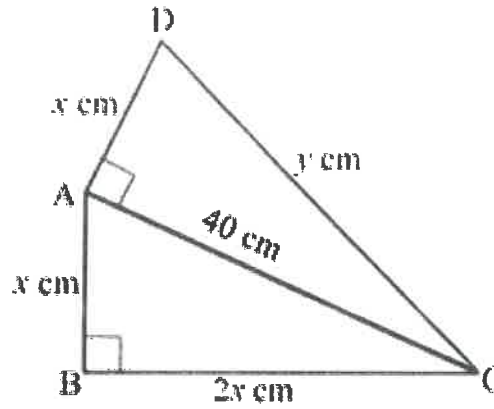
b)

Kevin is using a plank of wood that is 2.4 m long. How far away from the base of the veranda will the plank of wood sit, correct to two decimal places?

$$x^2 = 2.4^2 - 0.7^2 \checkmark$$

$$x = 2.30 \text{ m} /$$

2



- a Consider triangle ABC and find the exact value of x

$$\begin{aligned}(2x)^2 + x^2 &= 40^2 \quad \checkmark \\ 4x^2 + x^2 &= 1600 \quad \checkmark \\ 5x^2 &= 1600\end{aligned}$$

$$\begin{aligned}x &= \sqrt{\frac{1600}{5}} \quad \checkmark \\ x &= \sqrt{320} \quad \checkmark\end{aligned}$$

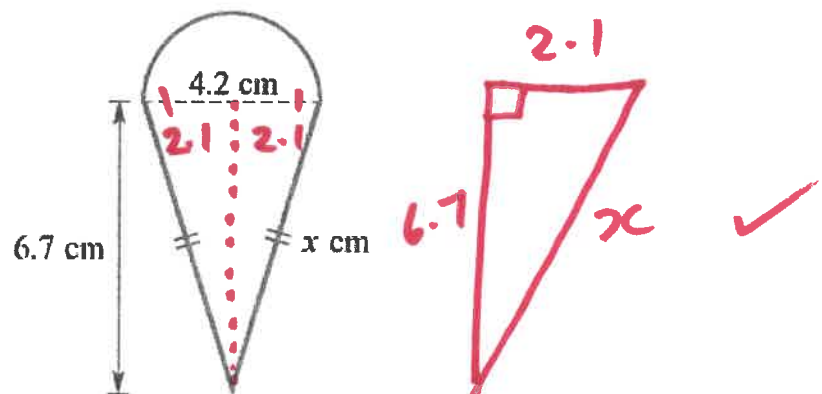
- b Use triangle ADC to find the value of y to 2 decimal places.

$$\begin{aligned}y^2 &= (\sqrt{320})^2 + 40^2 \quad \checkmark \\ y^2 &= 320 + 1600 \\ y &= \sqrt{1920} \\ y &= 43.82 \text{ cm} \quad \checkmark\end{aligned}$$

(5)

12

Consider the following composite shape.



- a Use Pythagoras' theorem to find the value of x , correct to one decimal place.

$$\begin{aligned}x^2 &= 6.7^2 + 2.1^2 \quad \checkmark \\ x &= 7.0 \text{ cm} \quad \checkmark\end{aligned}$$

(3)